

Arnav Bansal

arnavb9@illinois.edu | arnavbansal.dev | linkedin.com/in/arnavbansal11 | github.com/bansalarnav

EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science | GPA: 4.0/4.0

Champaign-Urbana, IL

Aug. 2025 – May 2029

Relevant Coursework: Data Structures, Algorithms, Computer Architecture, Probability & Statistics, Data Science

EXPERIENCE

Software Engineering Intern

Jun. 2026 – Aug. 2026

Open Science Stack

Delhi, India

- Built a **deterministic verification pipeline** for LLM-generated laboratory code, detecting **55+ classes of errors** before physical execution.
- Designed a two-stage validation system combining **static and dynamic analysis**, first through **language-server diagnostics**, and then a **stateful simulator**, exploring every reachable branch of code.

Undergraduate Research Assistant

May 2026 – Present

Distributed Protocols Research Group, UIUC

Urbana, IL

- Built a benchmark in **Go** to **benchmark LLM inference systems** across **seven workloads and ~250k requests**, including chat, RAG, long-context, code, and agentic traces, measuring **TTFT, latency, token throughput, and KV-cache behavior** under controlled prefix reuse and bursty traffic.
- Developed **agentrecap**, a tool to analyze local coding agent (Claude Code, Codex, Cursor, OpenCode etc.) sessions, studying cost, tool use, latency, token/cache efficiency, failures, and subagent behavior.

Software Engineering Intern

Mar. 2024 – Oct. 2024

InfoEdge India Pvt. Ltd.

Delhi, India

- Built a server-rendered career counseling platform with **React and Next.js**, including a context-aware AI assistant backed by persistent user history.
- Designed a **semantic summarization algorithm** that reduced prompt token usage by **80%** while preserving response quality and personalization.

PROJECTS

Alfie | TypeScript, Hono, Cloudflare, PostgreSQL, Redis

2026 – Present

- Built a **text-based AI personal assistant** with memory, web search, browser use, and durable scheduled tasks.
- Designed an actor-based main/worker agent architecture with message-passing using **Cloudflare Durable Objects** to keep conversations responsive while coordinating multiple **asynchronous, long-running tasks**.
- Implemented a **sandboxed, extensible plugin system** using **QuickJS**, enabling users to connect third-party services and AI agents to **create and modify integrations at runtime** on demand.

Railyard | Rust, Pingora, Podman

2026 – Present

- Building a **self-hosted deployment platform** that brings a Railway-style developer experience to any VPS through a **Rust daemon** and local CLI.
- Developed a **Pingora-based reverse-proxy**, with **automatic SSL termination** and service routing.
- Architecting a Podman orchestration engine supporting health-gated zero-downtime rollouts, automatic rollback, horizontal scaling, persistent volumes, scheduled jobs, and private service networking
- Built a secure, declarative control plane with **Ed25519-signed requests**, project-scoped access, single-use invitations, and a **typed manifest** enabling developers and **AI agents to manage deployments end-to-end**.

Cyclr | React Native, Bun/Elysia, PostgreSQL, ESP32

2025

- Built an end-to-end **bike-sharing platform**, including a React Native app, Bun/Elysia API, PostgreSQL backend, and custom IoT smart lock built with an **ESP32**.
- Implemented BLE/GATT service advertising and communication, and **HMAC-SHA256 challenge-response authentication** using device-generated nonces to securely control physical locks, preventing replay attacks.
- Won at UIUC's Technology Entrepreneur Center's Idea Fair

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Rust, Go, Python, C/C++, Dart, Kotlin, SQL

Libraries & Frameworks: React, Next.js, React Native, Hono, Elysia, Flutter, Pandas, NumPy

Systems & Tools: Linux, Git, Docker, BuildKit, containerd, Pingora, PostgreSQL, Redis, Cloudflare Workers, BLE/IoT